

# High-Performance Hydroponic Growing

Dragoon Mountains, Arizona

---

*Masterblend 4-18-38 · Calcium Nitrate · Modern LED Lighting  
DWC · Dutch Bucket · NFT · VPD · EC Management · Automation*

---

# Contents

---

|           |                                                                                            |
|-----------|--------------------------------------------------------------------------------------------|
| <b>1</b>  | <b>The Masterblend System</b><br>3-part formula, mixing ratios, stock solutions            |
| <b>2</b>  | <b>Water Quality &amp; EC Management</b><br>Source water, TDS, bicarbonates, target ranges |
| <b>3</b>  | <b>Modern LED Grow Lights</b><br>Spectrum, PPFD, DLI, positioning for your setup           |
| <b>4</b>  | <b>Growing Systems</b><br>DWC, Dutch bucket, NFT — setup and best uses                     |
| <b>5</b>  | <b>Environmental Control</b><br>VPD, temperature, CO2, airflow                             |
| <b>6</b>  | <b>Nutrient Schedules</b><br>Stage-by-stage feeding charts for key crops                   |
| <b>7</b>  | <b>pH &amp; EC Monitoring Protocol</b><br>Daily routines, drift management, adjustments    |
| <b>8</b>  | <b>Crop-Specific Optimization</b><br>Dialled-in settings for tomatoes, peppers, greens     |
| <b>9</b>  | <b>Automation &amp; Monitoring</b><br>Controllers, sensors, alerts — scaling up            |
| <b>10</b> | <b>Troubleshooting</b><br>Deficiency charts, EC problems, light issues                     |

1

The Masterblend System

3-part formula · mixing ratios · stock solutions

Masterblend 4-18-38 is a professional-grade, water-soluble fertiliser used by commercial greenhouse growers worldwide. Combined with calcium nitrate and magnesium sulfate, it delivers every essential macro and micronutrient in precise, plant-available form. Mastering the mixing protocol is the foundation of this system.

The 3-Part System

| Part | Product                        | NPK      | Primary Role                           | Mix Order         |
|------|--------------------------------|----------|----------------------------------------|-------------------|
| A    | Masterblend 4-18-38            | 4-18-38  | P, K, Mg + all micronutrients          | 1st               |
| B    | Magnesium Sulfate (Epsom Salt) | 0-0-0    | Mg, S — boosts Masterblend Mg          | 2nd               |
| C    | Calcium Nitrate 15.5-0-0       | 15.5-0-0 | Ca, N — never mix with A+B concentrate | 3rd — always last |

■ **Note:** Never mix calcium nitrate directly with Masterblend concentrate — they will react and precipitate, locking out nutrients. Always dissolve each part separately into the reservoir or in separate stock solutions.

Mixing Ratios

| Use Case                | Masterblend | Epsom Salt | Cal-Nitrate | Target EC | Per Volume |
|-------------------------|-------------|------------|-------------|-----------|------------|
| Seedlings / cuttings    | 1.2 g       | 0.6 g      | 1.2 g       | 0.8–1.0   | per gallon |
| Vegetative growth       | 2.4 g       | 1.2 g      | 2.4 g       | 1.4–1.8   | per gallon |
| Fruiting / flowering    | 3.0 g       | 1.5 g      | 3.0 g       | 2.0–2.4   | per gallon |
| Heavy fruiting (tomato) | 3.6 g       | 1.8 g      | 3.6 g       | 2.4–3.0   | per gallon |
| 5-gallon standard       | 12 g        | 6 g        | 12 g        | 2.0–2.4   | per 5 gal  |

Full Nutrient Profile at Standard Rate (per gallon)

| Macro          | ppm | Micro          | ppm | Micro           | ppm  |
|----------------|-----|----------------|-----|-----------------|------|
| Nitrogen (N)   | 116 | Iron (Fe)      | 2.4 | Zinc (Zn)       | 0.3  |
| Phosphorus (P) | 47  | Boron (B)      | 1.2 | Copper (Cu)     | 0.3  |
| Potassium (K)  | 188 | Manganese (Mn) | 1.2 | Molybdenum (Mo) | 0.06 |
| Calcium (Ca)   | 113 | Chloride (Cl)  | 12  |                 |      |
| Magnesium (Mg) | 41  | Sulfur (S)     | 58  |                 |      |

## Making Concentrated Stock Solutions

For larger systems or repeated mixing, making stock concentrates saves time and improves consistency. Keep Stock A and Stock B in separate containers — never combine them as concentrates.

| Stock   | Contents                         | Concentration                 | Dose Rate                 |
|---------|----------------------------------|-------------------------------|---------------------------|
| Stock A | Masterblend 4-18-38 + Epsom Salt | 1 lb each per gallon of water | 1 oz per gallon reservoir |
| Stock B | Calcium Nitrate 15.5-0-0 only    | 1 lb per gallon of water      | 1 oz per gallon reservoir |

**Pro tip:** Label stock bottles clearly with date mixed. Use within 2 weeks. Store in a cool, dark location to prevent degradation.

## 2

## Water Quality & EC Management

Source water · TDS baseline · bicarbonates · target ranges

SE Arizona well water is among the hardest in the US — high in calcium, magnesium, sodium, and bicarbonates. This directly affects how much Masterblend you add and how stable your pH will be. Knowing your baseline EC is not optional.

### Water Quality Targets

| Parameter    | Ideal Starting Water       | SE AZ Well Water (typical) | Action if Out of Range                                         |
|--------------|----------------------------|----------------------------|----------------------------------------------------------------|
| EC / TDS     | < 0.3 mS / < 150 ppm       | 0.3–0.8 mS / 200–500 ppm   | Reduce nutrient dose; consider RO filter or mix with rainwater |
| pH           | 6.5–7.0 (before treatment) | 7.5–8.5                    | Bicarbonate scrub + phosphoric or citric acid                  |
| Bicarbonates | < 50 ppm                   | 100–300 ppm                | Pre-acidify to neutralize before adding nutrients              |
| Calcium      | < 40 ppm                   | 60–150 ppm                 | Reduce calcium nitrate dose proportionally                     |
| Magnesium    | < 20 ppm                   | 20–50 ppm                  | Reduce Epsom salt dose proportionally                          |
| Sodium (Na)  | < 50 ppm                   | 10–80 ppm                  | High Na competes with K and Ca — flush regularly if > 80 ppm   |

### Bicarbonate Pre-Treatment Protocol

This is the most important step for Dragoon Mountain growers using well water. Bicarbonates buffer pH upward and fight every pH-down correction you make. Neutralize them first, then build your nutrient solution on clean water.

| Step | Action                                             | What to Observe                                                        |
|------|----------------------------------------------------|------------------------------------------------------------------------|
| 1    | Fill reservoir with source water                   | Note starting pH and EC                                                |
| 2    | Add phosphoric acid (pH Down) drop by drop         | Water will fizz — CO <sub>2</sub> releasing as bicarbonates neutralize |
| 3    | Wait for fizzing to completely stop                | This means bicarbonates are fully neutralized                          |
| 4    | Adjust pH to 5.8–6.0                               | Now pH is stable — won't bounce back                                   |
| 5    | Add Masterblend, then Epsom salt, then Cal-Nitrate | In this order, dissolved separately                                    |
| 6    | Final check: EC and pH                             | Adjust if needed, then begin growing                                   |

### Best Water Sources for This System

| Source               | Baseline EC | Suitability     | Recommendation                                                       |
|----------------------|-------------|-----------------|----------------------------------------------------------------------|
| RO filtered water    | ~0.0        | ★★★★★ Perfect   | Gold standard. Add nutrients from zero baseline. Most precise.       |
| Monsoon rainwater    | < 0.05      | ★★★★★ Excellent | Collect July–Sept. Store in sealed dark tanks. Nearly as good as RO. |
| RO/rain blend 50:50  | 0.1–0.2     | ★★★★ Very Good  | Stretches RO supply. Consistent and predictable.                     |
| Well water (treated) | 0.3–0.8     | ★★★ Usable      | Requires bicarbonate scrub. Reduce nutrient dose accordingly.        |
| Untreated well water | 0.3–0.8     | ★★ Difficult    | Inconsistent results. Invest in RO unit for best performance.        |

**Investment advice:** A basic RO (reverse osmosis) filter unit costs \$150–\$300 and is the single best investment for precision growing with Masterblend in this region. It eliminates all source water variables and makes your results completely reproducible.

3

Modern LED Grow Lights

Spectrum · PPFD · DLI · positioning

Modern quantum board LED grow lights have transformed indoor growing. They deliver full-spectrum light at high efficiency (2.5–3.0  $\mu\text{mol}/\text{J}$ ), produce far less heat than HPS, and can be tuned to the exact spectrum each crop needs. In the Dragoon Mountains, LEDs also allow year-round growing independent of season or weather.

Key Light Metrics

| Term           | What It Means                                                              | Why It Matters                                                                                                       |
|----------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| PPFD           | Photosynthetic Photon Flux Density ( $\mu\text{mol}/\text{m}^2/\text{s}$ ) | Measures instantaneous light intensity at canopy level. The most important number.                                   |
| DLI            | Daily Light Integral ( $\text{mol}/\text{m}^2/\text{day}$ )                | Total light received per day. $\text{PPFD} \times \text{hours} \div 3600$ . Target varies by crop.                   |
| PPE / Efficacy | $\mu\text{mol}$ of light per joule of electricity                          | Higher = more efficient. Quality LEDs: 2.5–3.0 $\mu\text{mol}/\text{J}$ . HPS: $\sim 1.7$ $\mu\text{mol}/\text{J}$ . |
| Spectrum       | Wavelength blend (nm) of the light output                                  | Full spectrum (3000K+5000K) for veg; add red (660nm) for flowering/fruiting.                                         |
| Photoperiod    | Hours of light per 24-hour cycle                                           | Veg: 18 hrs on / 6 off. Fruiting: 12–16 hrs depending on crop.                                                       |

PPFD Targets by Crop & Stage

| Crop / Stage                 | PPFD Target ( $\mu\text{mol}/\text{m}^2/\text{s}$ ) | DLI Target ( $\text{mol}/\text{m}^2/\text{day}$ ) | Photoperiod |
|------------------------------|-----------------------------------------------------|---------------------------------------------------|-------------|
| Seedlings / propagation      | 100 – 200                                           | 5 – 10                                            | 18 hrs      |
| Leafy greens (lettuce, kale) | 200 – 400                                           | 12 – 17                                           | 16–18 hrs   |
| Herbs (basil, cilantro)      | 300 – 500                                           | 15 – 20                                           | 16–18 hrs   |
| Tomato — vegetative          | 400 – 600                                           | 20 – 25                                           | 18 hrs      |
| Tomato — fruiting            | 600 – 900                                           | 25 – 35                                           | 14–16 hrs   |
| Pepper — vegetative          | 400 – 600                                           | 18 – 25                                           | 18 hrs      |
| Pepper — fruiting            | 600 – 800                                           | 22 – 30                                           | 14–16 hrs   |
| Cucumber — fruiting          | 700 – 1000                                          | 30 – 40                                           | 14–16 hrs   |

Recommended LED Models (2025)

| Light                 | Power | Coverage | Efficacy                      | Best For                            |
|-----------------------|-------|----------|-------------------------------|-------------------------------------|
| Spider Farmer SF-4000 | 400W  | 4×4 ft   | 2.9 $\mu\text{mol}/\text{J}$  | Tomatoes, peppers, high-light crops |
| Mars Hydro FC-E4800   | 480W  | 4×4 ft   | 2.85 $\mu\text{mol}/\text{J}$ | Heavy fruiting crops                |

|                       |      |        |            |                              |
|-----------------------|------|--------|------------|------------------------------|
| Gavita Pro 1700e LED  | 645W | 5×5 ft | 2.6 μmol/J | Commercial-scale fruiting    |
| Spider Farmer SE-3000 | 300W | 3×4 ft | 2.7 μmol/J | Mixed veg + herb production  |
| HLG 100 V2 (QB)       | 100W | 2×2 ft | 2.7 μmol/J | Seedlings, small herb setups |

### Dragoon Mountains Advantage: Supplemental Lighting

At 4,500–7,500 ft elevation, natural sunlight intensity is 15–20% higher than at sea level. For greenhouse or semi-outdoor growing, this means:

- Natural DLI during fall/spring can reach 40–50 mol/m²/day — more than enough for any crop
- LEDs are most valuable in winter (low sun angle) and for extending photoperiod after sunset
- Supplemental LED lighting in a greenhouse — just 2–4 hrs per day — dramatically boosts winter yields
- Use shade cloth AND LEDs: shade for midday UV protection, LEDs for early morning / evening extension



4

Growing Systems

DWC · Dutch Bucket · NFT · setup guides

Three systems stand out as best suited for high-performance growing in the Dragoon Mountains: Deep Water Culture for leafy crops and smaller fruiting plants, Dutch Bucket for large fruiting crops, and NFT for greens at scale. Each pairs excellently with the Masterblend nutrient system.

System Comparison

| System                   | Best Crops                         | Complexity | Water Use      | Masterblend Compatibility |
|--------------------------|------------------------------------|------------|----------------|---------------------------|
| Deep Water Culture (DWC) | Tomato, pepper, leafy greens       | Low        | Very efficient | ★★★★★ Ideal               |
| Dutch Bucket (Bato)      | Tomato, cucumber, pepper, eggplant | Medium     | Efficient      | ★★★★★ Ideal               |
| NFT (Nutrient Film)      | Lettuce, herbs, strawberries       | Medium     | Most efficient | ★★★★★ Ideal               |
| Kratky (passive DWC)     | Lettuce, herbs, greens             | Very Low   | Efficient      | ★★★★ Very Good            |
| Ebb & Flow               | Most crops                         | Medium     | Moderate       | ★★★★ Good                 |

Deep Water Culture (DWC) — Setup Guide

| Component      | Specification / Note                                                          |
|----------------|-------------------------------------------------------------------------------|
| Reservoir      | 5–17 gallon dark bucket or tote. Insulate or bury for temp control.           |
| Air pump       | Minimum 1W per gallon of reservoir volume. 24/7 operation.                    |
| Air stone      | Medium or large — maximise dissolved oxygen in root zone.                     |
| Net pots       | 2" or 3" net pots with hydroton or perlite to anchor plants.                  |
| Water level    | Initially roots just touching solution. Drop 1–2" as roots develop.           |
| Reservoir temp | Keep 65–72°F — critical. Higher temps = root rot and low O2.                  |
| Res change     | Full change every 7–14 days for fruiting crops; top off with fresh mix daily. |

Dutch Bucket System — Setup Guide

| Component     | Specification / Note                                        |
|---------------|-------------------------------------------------------------|
| Buckets       | 5-gallon Dutch buckets or Bato buckets. 1 plant per bucket. |
| Media         | 70% perlite / 30% hydroton. Rinse thoroughly before use.    |
| Drip emitters | 1 GPH emitter per bucket. Timer: 4–6 cycles per day.        |

|                      |                                                                             |
|----------------------|-----------------------------------------------------------------------------|
| <b>Drain line</b>    | Slope ¼" per foot back to reservoir. Single return line.                    |
| <b>Reservoir</b>     | 50–100 gal for 8–16 bucket system. Insulate from heat.                      |
| <b>Recovery</b>      | Recirculating system — collect and reuse drain water.                       |
| <b>EC management</b> | Check reservoir EC daily. Top off with half-strength mix as water depletes. |

**Dutch Bucket advantage:** Dutch Bucket with perlite is the #1 choice for tomatoes and peppers in controlled environments. It gives roots excellent oxygen, drains perfectly, and allows individual plant EC/pH management.

## 5

## Environmental Control

VPD · temperature · CO2 · airflow

Nutrients and lights matter, but environment drives the ceiling on your yields. VPD (Vapour Pressure Deficit) is the single most overlooked variable in small-scale growing. Getting it right unlocks faster growth, better nutrient uptake, and dramatically healthier plants.

## VPD — Vapour Pressure Deficit

VPD measures the difference between how much moisture the air holds and how much it could hold. Too low = sluggish transpiration, wet leaves, fungal risk. Too high = plants close stomata, nutrient uptake stops, wilting.

| Growth Stage             | Temperature       | Humidity | Target VPD (kPa) |
|--------------------------|-------------------|----------|------------------|
| Seedlings / clones       | 72–77°F (22–25°C) | 70–80%   | 0.4 – 0.8        |
| Early vegetative         | 72–79°F (22–26°C) | 60–70%   | 0.8 – 1.0        |
| Late vegetative          | 75–82°F (24–28°C) | 55–65%   | 1.0 – 1.2        |
| Flowering / fruiting     | 68–79°F (20–26°C) | 45–55%   | 1.2 – 1.6        |
| Late fruiting / ripening | 65–75°F (18–24°C) | 40–50%   | 1.4 – 1.8        |

## Dragoon Mountains Climate Interaction

- **Monsoon season (July–Sept):** Outdoor humidity 50–70% — VPD drops naturally. Reduce watering frequency. Watch for botrytis and powdery mildew.
- **Dry season (Oct–June):** Humidity often 10–30% — VPD is very high. Add a humidifier to your grow space or plants will constantly stress.
- **High elevation UV:** If using a greenhouse with natural light, UV intensity is 15–20% higher than sea level — accelerates leaf surface drying.

## Temperature & CO2

| Parameter             | Optimal Range     | Dragoon-Specific Note                                                                                       |
|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------|
| Air temp (lights on)  | 72–82°F (22–28°C) | Summer nights naturally cool — great for fruiting crops requiring cold nights                               |
| Air temp (lights off) | 62–70°F (17–21°C) | Dragoon night temps often ideal naturally — use outdoor air exchange at night                               |
| Root zone temp        | 65–72°F (18–22°C) | Most critical parameter. Bury/insulate reservoirs. Higher = root rot.                                       |
| CO2 (ambient)         | 400 ppm (natural) | At elevation, CO2 is slightly lower — sealed indoor grow rooms benefit from enrichment                      |
| CO2 (enriched)        | 800–1200 ppm      | Increases photosynthesis 20–30%. Only effective with high PPFD (600+ $\mu\text{mol}/\text{m}^2/\text{s}$ ). |

|         |                          |                                                                                    |
|---------|--------------------------|------------------------------------------------------------------------------------|
| Airflow | Gentle constant movement | Strengthens stems, prevents hot spots, reduces fungal risk during monsoon humidity |
|---------|--------------------------|------------------------------------------------------------------------------------|

## 6

## Nutrient Schedules

Stage-by-stage feeding charts for key crops

Masterblend's simplicity is one of its greatest strengths — the same 2:1:2 ratio (Masterblend : Epsom : Cal-Nitrate) scales linearly across the entire growth cycle. You adjust concentration, not the ratio. The charts below are per-gallon working solution.

## Tomato — Full Cycle Schedule

| Week                | Stage                 | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|---------------------|-----------------------|------------|---------------|---------------|-----------|---------|
| 1–2                 | Germination           | —          | —             | —             | 0.0       | 6.0     |
| 2–4                 | Seedling              | 1.2        | 0.6           | 1.2           | 0.8–1.0   | 5.8–6.0 |
| 4–7                 | Early veg             | 2.4        | 1.2           | 2.4           | 1.4–1.8   | 5.8–6.0 |
| 7–10                | Late veg / transition | 2.8        | 1.4           | 2.8           | 1.8–2.2   | 5.8–6.2 |
| 10–14               | First truss flowering | 3.0        | 1.5           | 3.0           | 2.0–2.4   | 5.8–6.2 |
| 14–20               | Heavy fruiting        | 3.6        | 1.8           | 3.6           | 2.4–3.0   | 6.0–6.2 |
| Final<br>1–2<br>wks | Flush / finish        | 0.5        | 0.3           | 0.5           | 0.5–0.8   | 6.0     |

## Pepper — Full Cycle Schedule

| Week | Stage      | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|------|------------|------------|---------------|---------------|-----------|---------|
| 1–3  | Seedling   | 1.2        | 0.6           | 1.2           | 0.8–1.0   | 5.8–6.0 |
| 3–6  | Vegetative | 2.4        | 1.2           | 2.4           | 1.4–1.8   | 5.8–6.0 |
| 6–10 | Pre-flower | 2.8        | 1.4           | 2.8           | 1.8–2.2   | 5.8–6.2 |
| 10+  | Fruiting   | 3.0        | 1.5           | 3.0           | 2.2–2.8   | 6.0–6.2 |

## Leafy Greens &amp; Herbs — Schedule

| Crop    | Stage      | MB (g/gal) | Epsom (g/gal) | Cal-N (g/gal) | Target EC | pH      |
|---------|------------|------------|---------------|---------------|-----------|---------|
| Lettuce | All stages | 1.2–1.8    | 0.6–0.9       | 1.2–1.8       | 0.8–1.4   | 5.8–6.2 |
| Basil   | Vegetative | 2.0        | 1.0           | 2.0           | 1.4–1.8   | 5.8–6.0 |
| Kale    | All stages | 2.0–2.4    | 1.0–1.2       | 2.0–2.4       | 1.4–2.0   | 5.8–6.2 |
| Spinach | All stages | 1.6        | 0.8           | 1.6           | 1.0–1.6   | 6.0–6.5 |

|          |            |         |         |         |         |         |
|----------|------------|---------|---------|---------|---------|---------|
| Cilantro | All stages | 1.2–1.6 | 0.6–0.8 | 1.2–1.6 | 0.8–1.4 | 6.0–6.5 |
|----------|------------|---------|---------|---------|---------|---------|

■ **Note:** These are starting points. Always measure your actual reservoir EC and adjust. If plants show tip burn, lower EC. If growth is slow and leaves are pale, raise EC.

## 7

## pH &amp; EC Monitoring Protocol

Daily routines · drift management · adjustments

Consistent monitoring is what separates good results from great ones. With Masterblend and hard desert water, daily checks for the first two weeks of any new reservoir will reveal your water's behaviour and let you build a predictable maintenance routine.

## Daily Monitoring Routine

| Frequency | Task                  | Notes                                                          |
|-----------|-----------------------|----------------------------------------------------------------|
| Morning   | pH check              | Adjust if outside 5.8–6.2. Note drift direction and amount.    |
| Morning   | EC check              | Top off with half-strength mix if EC dropped from evaporation. |
| Morning   | Water level           | Top off reservoirs. Never let roots dry out.                   |
| Evening   | Visual plant check    | Look for leaf colour, wilting, pest signs, root colour.        |
| Weekly    | Full reservoir change | Fruiting crops: change every 7–10 days. Greens: every 14 days. |
| Weekly    | Meter calibration     | Calibrate pH meter with buffer solutions (pH 4.0 and 7.0).     |
| Weekly    | Root inspection       | Healthy roots = bright white. Brown/slippery = problem.        |
| Monthly   | System flush          | Flush lines and reservoir with plain water between cycles.     |

## pH Adjustment — Precision Protocol

| Adjustment        | Product                          | Application                                                 | Dragoon Local Option                                    |
|-------------------|----------------------------------|-------------------------------------------------------------|---------------------------------------------------------|
| pH Down           | Phosphoric acid 85% (GH pH Down) | Add 1 ml per 10 gal, wait 10 min, retest                    | Fermented prickly pear vinegar for minor adjustments    |
| pH Up             | Potassium hydroxide (GH pH Up)   | Add 1 ml per 10 gal, wait 10 min, retest                    | Dilute oak ash lye solution for occasional upward drift |
| Bicarbonate flush | Citric acid or phosphoric acid   | Pre-treat water until fizzing stops before adding nutrients | Fermented citric acid from local fruit fermentation     |

## Understanding EC Drift

- **EC rising:** Plants are drinking more water than nutrients — top off with plain RO/rain water, not full-strength mix
- **EC falling:** Plants are consuming nutrients faster than water — top off with half-strength Masterblend mix
- **EC stable, pH rising:** Classic bicarbonate effect from hard water — pre-treat next reservoir change with acid before adding nutrients
- **EC stable, pH falling:** Biological activity or ammonium uptake — normal in warm weather; adjust up with pH Up

## 8

## Crop-Specific Optimization

Dialled-in settings for top crops

## Indeterminate Tomato

| Parameter      | Setting / Note                                                                               |
|----------------|----------------------------------------------------------------------------------------------|
| System         | Dutch Bucket (perlite) or DWC                                                                |
| EC             | 2.0–3.0 mS/cm                                                                                |
| pH             | 5.8–6.2                                                                                      |
| Light (PPFD)   | 600–900 $\mu\text{mol}/\text{m}^2/\text{s}$                                                  |
| DLI            | 25–35 $\text{mol}/\text{m}^2/\text{day}$                                                     |
| Temp day/night | 75–82°F / 62–68°F                                                                            |
| Photoperiod    | 14–16 hrs                                                                                    |
| VPD fruiting   | 1.2–1.6 kPa                                                                                  |
| Key tip        | Calcium is critical — never skip Cal-Nitrate. Blossom end rot = Ca deficiency or pH lockout. |

## Chile &amp; Bell Pepper

| Parameter      | Setting / Note                                                                                               |
|----------------|--------------------------------------------------------------------------------------------------------------|
| System         | Dutch Bucket or DWC                                                                                          |
| EC             | 1.8–2.8 mS/cm                                                                                                |
| pH             | 5.8–6.2                                                                                                      |
| Light (PPFD)   | 600–800 $\mu\text{mol}/\text{m}^2/\text{s}$                                                                  |
| DLI            | 22–30 $\text{mol}/\text{m}^2/\text{day}$                                                                     |
| Temp day/night | 72–82°F / 60–65°F                                                                                            |
| Photoperiod    | 14–16 hrs                                                                                                    |
| VPD fruiting   | 1.2–1.6 kPa                                                                                                  |
| Key tip        | Peppers are heavy Mg users — double the Epsom salt dose to 2.4 g/gal (vs 1.2 standard) once fruiting begins. |



## Lettuce & Leafy Greens

| Parameter    | Setting / Note                                                                                                                    |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| System       | NFT or Kratky                                                                                                                     |
| EC           | 0.8–1.6 mS/cm                                                                                                                     |
| pH           | 5.8–6.5                                                                                                                           |
| Light (PPFD) | 200–400 $\mu\text{mol}/\text{m}^2/\text{s}$                                                                                       |
| DLI          | 12–17 $\text{mol}/\text{m}^2/\text{day}$                                                                                          |
| Temp         | 60–72°F — cool is better                                                                                                          |
| Photoperiod  | 16–18 hrs                                                                                                                         |
| VPD          | 0.8–1.2 kPa                                                                                                                       |
| Key tip      | Lower EC produces tender, mild leaves. High EC causes tip burn and bitterness. In summer, grow in the coolest part of your space. |

## Basil

| Parameter    | Setting / Note                                                                                                                         |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| System       | DWC or NFT                                                                                                                             |
| EC           | 1.4–2.0 mS/cm                                                                                                                          |
| pH           | 5.8–6.2                                                                                                                                |
| Light (PPFD) | 300–500 $\mu\text{mol}/\text{m}^2/\text{s}$                                                                                            |
| DLI          | 15–20 $\text{mol}/\text{m}^2/\text{day}$                                                                                               |
| Temp         | 70–85°F — loves heat                                                                                                                   |
| Photoperiod  | 16–18 hrs                                                                                                                              |
| VPD          | 1.0–1.4 kPa                                                                                                                            |
| Key tip      | Harvest frequently to prevent bolting. High PPFD produces more essential oils and stronger flavour. Ideal Dragoon monsoon season crop. |

## 9

## Automation &amp; Monitoring

Controllers · sensors · alerts · scaling up

Automation transforms a high-maintenance daily task into a manageable weekly check-in. For Dragoon Mountain growers — where temperature swings, monsoon humidity changes, and remote location can make daily attention difficult — even basic automation is a major quality-of-life improvement.

## Essential Automation — Priority Order

| Device                                     | Function                                       | Cost Range | Priority                    |
|--------------------------------------------|------------------------------------------------|------------|-----------------------------|
| pH/EC meter (digital)                      | Manual daily monitoring                        | \$30–150   | ★★★★★ Essential             |
| Digital timer (light cycle)                | Automates photoperiod — lights on/off          | \$15–30    | ★★★★★ Essential             |
| Ink-bird temp controller                   | Triggers fans/heaters at set temps             | \$25–50    | ★★★★★ Essential             |
| Digital hygrometer                         | Monitors temp + humidity for VPD               | \$15–25    | ★★★★ Important              |
| Dosatron / auto-doser                      | Automatically doses nutrients at set EC        | \$150–500  | ★★★ Recommended at scale    |
| Autopilot CO2 controller                   | Maintains CO2 at target ppm                    | \$150–300  | ★★★ If using CO2 enrichment |
| WiFi grow controller (Inkbird, Trolmaster) | Remote monitoring, alerts on phone             | \$100–400  | ★★★ Remote location value   |
| Auto top-off system                        | Maintains reservoir water level automatically  | \$50–150   | ★★★ Saves daily labour      |
| EC/pH dosing controller (Bluelab, Atlas)   | Continuously monitors and auto-doses nutrients | \$500–1500 | ★★ Advanced / commercial    |

## Dragoon-Specific Automation Priorities

- **Temperature alerts:** Large day/night swings mean reservoir temps can fluctuate significantly — a temp controller with phone alert is high value here
- **Humidity alerts during monsoon:** Rapid humidity swings July–September can push VPD out of range fast — a WiFi hygrometer with alerts is worth it
- **Remote monitoring:** If you're off-grid or travel frequently, WiFi sensors that alert your phone when pH or temp goes out of range prevent crop losses
- **Light timers:** Non-negotiable — consistent photoperiod is critical for fruiting triggers and preventing premature bolting in greens

**Solar note:** The Dragoon Mountains receive 300+ sunny days per year. A modest solar + battery setup (400W panels + 100Ah lithium battery) can power LED lights, pumps, and all monitoring equipment — making this system effectively off-grid.

## 10

## Troubleshooting

Deficiency charts · EC problems · light &amp; environment issues

## Nutrient Deficiency Quick Reference

| Symptom                                        | Likely Deficiency             | Masterblend Fix                                             | pH Range Cause                                 |
|------------------------------------------------|-------------------------------|-------------------------------------------------------------|------------------------------------------------|
| Yellow between leaf veins — older leaves first | Magnesium (Mg)                | Double Epsom salt dose (2.4 g/gal) for 1–2 weeks            | Locked out below 5.5                           |
| Yellow between leaf veins — new leaves first   | Iron (Fe) or Manganese (Mn)   | Check pH — most common cause. Adjust to 5.8–6.0             | Locked out above 6.5                           |
| Brown/dark tips, necrotic leaf edges           | Calcium (Ca) or potassium (K) | Check Cal-Nitrate was mixed last. Increase to 3.0 g/gal     | Locked out below 5.5 or above 7.0              |
| Blossom end rot on tomatoes                    | Calcium (Ca) deficiency       | Raise Cal-Nitrate to 3.6 g/gal. Check EC not too high.      | pH out of range or EC too high blocking uptake |
| Purple/reddish leaf undersides                 | Phosphorus (P)                | Check pH — P locks out below 5.0. Warm to 70°F+.            | Locked out below 5.0 or above 7.5              |
| Overall pale, slow growth                      | Nitrogen (N) — too low EC     | Increase overall Masterblend dose by 0.5 g/gal              | Check EC < 1.0 for fruiting crops              |
| Dark green leaves, stocky but slow             | Nitrogen (N) — too high EC    | Dilute reservoir. Do partial water change with fresh water. | EC above 3.5 in most cases                     |

## System &amp; Environment Problems

| Problem                                   | Cause                                               | Solution                                                                      |
|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------|
| Root rot (brown, slimy roots)             | Reservoir temp > 72°F or insufficient O2            | Insulate reservoir. Add more airstones. Add beneficial bacteria (Hydroguard). |
| White mineral crust on pots / media       | Salt buildup from hard water                        | Flush with RO/rain water. Switch primary water source. Check EC baseline.     |
| pH unstable — rises rapidly               | Bicarbonates in source water                        | Pre-acidify water before adding nutrients. Switch to RO or rainwater.         |
| Slow growth despite good EC/pH            | VPD out of range or root zone too warm              | Check temp and humidity. Verify root zone < 72°F. Check for root issues.      |
| Powdery mildew on leaves                  | High humidity + low airflow (common during monsoon) | Increase airflow. Reduce humidity below 60%. Remove affected leaves.          |
| Light burn — bleached/yellow top leaves   | LED too close or PPFD too high                      | Raise light height. Reduce intensity. Check manufacturer's height guide.      |
| Stretching / etiolation — long thin stems | PPFD too low / not enough light                     | Lower LED height. Increase intensity. Extend photoperiod to 18 hrs.           |

|                    |                              |                                                                      |
|--------------------|------------------------------|----------------------------------------------------------------------|
| Algae in reservoir | Light getting into reservoir | Cover all reservoir surfaces completely. Use opaque containers only. |
|--------------------|------------------------------|----------------------------------------------------------------------|

*With Masterblend's precision nutrition, modern high-efficacy LED lighting, and controlled-environment techniques, the Dragoon Mountains become an exceptional growing location — cooler summers than the low desert, abundant solar energy, monsoon rainwater, and long clear fall seasons all work in your favour.*